

Software Design Specification

Dalitz Plot Interactive Web Visualizer

1. Vision

Create an interactive web platform for building, exploring and visualizing three-body decay amplitude models. The application targets both education and amplitude-analysis research.

2. Objectives

Automatically validate decay channels using particles and qrules; suggest intermediate resonances; allow unrestricted custom models; update all observables reactively; generate weighted Monte Carlo samples; export reproducible configurations.

3. Functional Requirements

- Reactive interface with no Update buttons.
- Automatic decay validation after particle selection.
- Automatic Dalitz boundary calculation.
- Automatic resonance suggestions from qrules.
- Manual addition of arbitrary resonances.
- Editable masses, widths, spins, lineshapes, magnitudes and phases.
- Interactive Dalitz and invariant-mass projections.
- Weighted MC generation ($\leq 1\text{M}$ events).
- CSV export; future ROOT/Parquet.

4. Non-functional Requirements

Modular, extensible, reproducible, responsive, physics separated from UI, documented API, unit-tested numerical core.

5. System Architecture

Frontend: React + TypeScript + Plotly. Backend: FastAPI. Physics engine: particles, qrules, NumPy, SciPy. Backend exposes JSON APIs consumed by the frontend.

6. Physics Engine

Core classes: DecayModel, Resonance, LineShape, SpinFactor, BlattWeisskopf, AmplitudeTerm, AmplitudeModel, DalitzCalculator, MCGenerator. Each module is independent and replaceable.

7. Reactive Workflow

Selecting particles triggers validation and kinematic calculation. Adding/removing resonances rebuilds the model. Editing any parameter recomputes amplitudes, intensities, Dalitz plots, projections and derived quantities automatically.

8. Resonance Management

Suggested resonances come from qrules. Users may also search the particle database or define fully custom resonances. The application warns about unsupported hypotheses but never blocks them (Suggest, don't restrict).

9. Visualization

Dalitz plot, m_{12}^2 , m_{13}^2 and m_{23}^2 projections, resonance overlays, interference maps, phase maps, Argand diagrams and fit fractions.

10. Monte Carlo

Generate weighted phase-space events, compute weights from the current amplitude model and export four-momenta, invariant masses and weights.

11. JSON Model

Entire models are serializable, allowing save/load/share through JSON and eventually URLs.

12. Development Roadmap

Phase 1: architecture. Phase 2: decay validation. Phase 3: amplitude engine. Phase 4: visualization. Phase 5: Monte Carlo. Phase 6: advanced models (GS, Flatte, K-matrix, LASS).

13. Guiding Principles

Suggest, don't restrict. Everything is editable. Everything is reactive. Physics is independent from presentation.